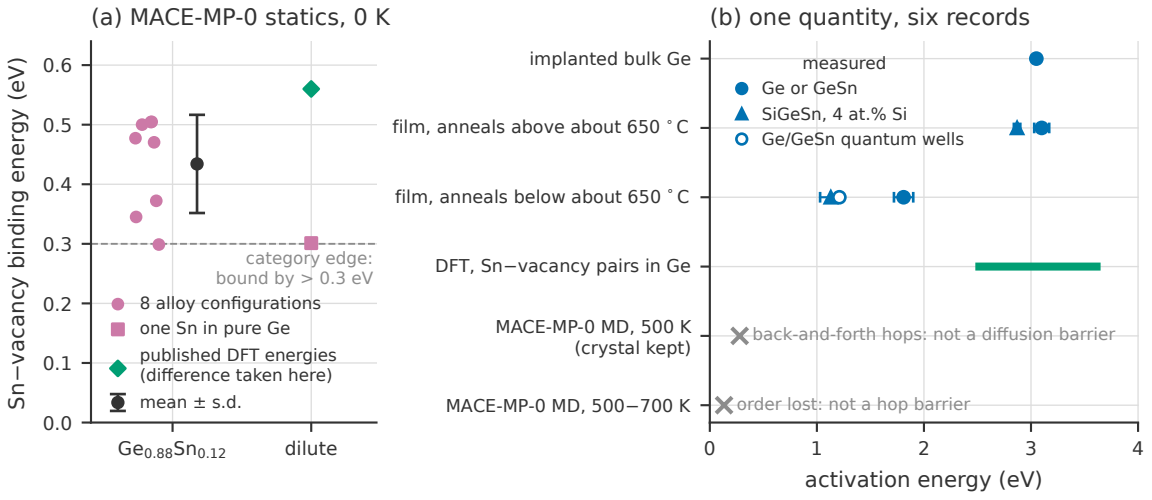
(a) MACE-MP-0 statics, 0 K
Sn−vacancy binding energy (eV)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
category edge:
bound by > 0.3 eV
8 alloy configurations
one Sn in pure Ge
published DFT energies
(difference taken here)
mean ± s.d.
Ge0.88Sn0.12
dilute
(b) one quantity, six records
implanted bulk Ge
film, anneals above about 650 °C
film, anneals below about 650 °C
DFT, Sn−vacancy pairs in Ge
MACE-MP-0 MD, 500 K
(crystal kept)
MACE-MP-0 MD, 500−700 K
measured
Ge or GeSn
SiGeSn, 4 at.% Si
Ge/GeSn quantum wells
back-and-forth hops: not a diffusion barrier
order lost: not a hop barrier
0
1
2
3
4
activation energy (eV)